3.1 Meaning and characteristics of qualitative research

Qualitative research is a methodological approach used to explore and understand phenomena in-depth by examining the underlying **reasons, motivations, and behaviors** of individuals. Unlike quantitative research, which focuses on numerical data and statistical analysis, qualitative research emphasizes gathering rich, descriptive data through methods such as interviews, observations, and analysis of textual or visual materials.

These are some of the most common qualitative methods: Observations: recording what you have seen, heard, or encountered in detailed field notes. Interviews: personally, asking people questions in one-on-one conversations. Focus groups: asking questions and generating discussion among a group of people.

Example

Here's an example of a qualitative research study:

Title: "**Exploring** the Experience of Transitioning to Remote Work During the COVID-19 Pandemic: A Qualitative Study"

Research Question: What are the experiences and challenges faced by employees transitioning to remote work during the COVID-19 pandemic?

Methodology:

1. Sampling: The researchers select a diverse sample of employees from various industries and professions who have recently transitioned to remote work due to the COVID-19 pandemic.
2. Data Collection: a. In-depth Interviews: Researchers conduct semi-structured interviews with the participants to explore their experiences, perceptions, and challenges related to remote work. b. Open-ended Surveys: Participants are also asked to complete open-ended survey questions to provide additional insights into their remote work experiences. c. Document Analysis: Any relevant documents such as emails, memos, or personal reflections related to the transition to remote work are collected and analyzed.
3. Data Analysis: Thematic Analysis: Researchers transcribe and analyze the interview and survey data to identify recurring themes and patterns related to the participants' experiences and challenges. b. Constant Comparison: The researchers compare data across different participants to identify similarities and differences in their experiences. c. Member Checking: Researchers may also validate their findings by returning to participants with the preliminary analysis to ensure accuracy and authenticity.
4. Findings: The researchers identify several key themes emerging from the data, including:
    - Challenges of establishing work-life balance
    - Communication difficulties and feelings of isolation
    - Technological barriers and access to resources
    - Adaptation strategies and coping mechanisms
5. Conclusion and Implications: The study concludes by discussing the implications of the findings for organizations and policymakers in supporting employees' transition to remote work, especially in the context of ongoing public health crises like the COVID-19 pandemic.

3.2 Purposive sampling

Purposive sampling, also known as **judgmental sampling** or **selective sampling**, is a non-probability sampling technique used in research studies. In contrast to random sampling, where participants are chosen by chance, purposive sampling relies on the researcher's judgment to select participants who best suit the research goals.

Also called judgmental sampling, this sampling method relies on the researcher’s judgment when identifying and selecting the individuals, cases, or events that can provide the best information to achieve the study’s objectives.

Purposive sampling is common in qualitative research and mixed methods research. It is particularly useful if you need to find information-rich cases or make the most out of limited resources, but is at high risk for research biases like observer bias.

Principles of purposeful sampling

There are several important principles of purposive sampling that you should consider when using this approach in your research studies:

1. **Clearly defined purpose -** The purpose of the study should be clearly defined, and the sample should be selected based on the characteristics or attributes that you're interested in studying.
2. **Representative sample -** The sample should be representative of the characteristics or attributes being studied.
3. **Bias -** Biases can come into play when anything other than random sampling is used, so be aware of any potential biases and take steps to minimize them.
4. **Expertise -** Having expertise in the topic being studied is an important part of sample selection. Without a solid understanding of the characteristics being selected, the population might not be as representative as it should be.

How is purposive sampling conducted?

The steps to conducting a study using purposive sampling will vary depending on the topic and preferences of the researchers involved. The five steps of purposive sampling as a general framework are:

1. Define the purpose of the study
2. Identify the sample of individuals or units
3. Obtain informed consent from individuals
4. Collect the data using appropriate research methods
5. Analyze the data

Common Types of Purposive Sampling:

Researchers can use several different types of purposive sampling methods, depending on what they're interested in studying and the specific research question they are trying to answer. In the list below, we'll discuss the various types of purposive sampling methods and provide examples of when each method might be used in research.

Maximum variation sampling

Maximum variation sampling involves selecting a sample of individuals or units representing the maximum range of variation within the characteristics or attributes the researcher is interested in studying. This type of sampling is used to understand the widest possible diversity of experiences or viewpoints within the population.

Homogeneous sampling

Homogeneous sampling involves selecting what is often a narrower sample of individuals or units that are similar or have the same characteristics or attributes. This type of sampling is used to study a specific subgroup within the population in depth.

Typical case sampling

Typical case sampling involves selecting a sample of individuals or units that are representative of the typical experiences or characteristics of the population. This type of sampling is used to understand the most common or average experiences or characteristics within the population.

Extreme/deviant case sampling

Extreme case sampling involves selecting a sample of individuals or units that are considered extreme or unusual in the characteristics or attributes the researcher is interested in studying. This type of sampling is used to understand unusual or exceptional experiences or characteristics within the population and are often viewed as outliers in a wider population.

Critical case sampling

Critical case sampling involves selecting a sample of individuals or units that are important or central to the research question or the population being studied. This type of sampling is used to understand key experiences or characteristics within the population.

Expert sampling

Expert sampling involves selecting a sample of individuals or units that have specialized knowledge or expertise in the topic or issue being studied. This type of sampling is used to gather insights and understanding from experts in the field, which can be used to develop follow-up studies.

Snowball Sampling: This technique is useful when studying hard-to-reach populations. Researchers start with a few initial participants and ask them to identify others who share the desired characteristics.

Advantages of Purposive Sampling:

- **Information-Rich Data:** By selecting participants with relevant experiences, purposive sampling often yields rich and detailed data that can provide deep insights.
- **Cost-Effective:** Since purposive sampling focuses on specific groups, it can be more time- and cost-effective than random sampling methods that require larger sample sizes.
- **Feasible for Specific Populations:** For studies of rare populations or those where random sampling is impractical, purposive sampling can be a viable option.

Disadvantages of Purposive Sampling:

- **Selection Bias:** Because researchers choose participants based on judgment, there's a risk of bias. The chosen sample may not accurately reflect the broader population.
- **Limited Generalizability:** Findings from purposive samples are generally not generalizable to the entire population, as the sample may not be representative.
- **Unobservable Characteristics:** There's a possibility of excluding participants with relevant experiences or characteristics that the researcher may not have foreseen.

Purposive sampling is a valuable tool for qualitative research and studies where in-depth understanding of a specific group or phenomenon is desired. However, it's crucial to be aware of the limitations and potential biases associated with this technique.

3.3 Data collection strategies: semi-structured and unstructured interview, participant observation, focus group discussion

Qualitative research thrives on rich, descriptive data that unlocks the "why" and "how" behind human experiences. Here's an in-depth exploration of four commonly used data collection strategies, along with examples to illustrate their application:

1. Semi-structured Interviews: A Guided Exploration

Imagine a conversation with a purpose. Semi-structured interviews use a pre-defined interview guide with key questions and areas to explore., there's flexibility to delve deeper based on the participant's unique responses are called as un-structural. This allows for:

- **Focused Data:** The interview guide ensures you gather relevant information on your research topic.
- **In-depth Exploration:** Follow-up questions and probes based on the participant's answers can reveal unexpected insights and nuances.
- **Consistency with Flexibility:** The guide provides a framework for comparison across interviews, while allowing for individual experiences to shine through.

Example: You're researching the impact of social media on teenagers' self-esteem. Your interview guide might include questions like:

* How often do you use social media platforms?
* Can you describe how social media makes you feel about yourself?
* (Follow-up based on response) Do you ever compare yourself to others on social media? How does that make you feel?

2. Unstructured Interviews: Unveiling the Unexpected

Think of an open-ended conversation where the participant takes the lead. Unstructured interviews have minimal structure, often starting with a broad question like "Tell me about your experiences with online learning." The researcher then actively listens and prompts the participant to elaborate, allowing the discussion to flow organically. This approach is valuable for:

- **Deep Exploration:** Unforeseen topics and unique perspectives can emerge organically.
- **Freedom of Expression:** Participants feel comfortable sharing their experiences and stories in their own words.
- **Exploratory Research:** This method is ideal when you're starting with a broad research question and need to understand the landscape before defining specific areas of focus.

Example: You're researching the challenges faced by first-generation college students. You might begin with a question like: "Can you walk me through your experience as a first-generation college student?" This could lead to discussions about financial aid, navigating campus life, or feelings of isolation.

3. Participant Observation: Immersing Yourself in the Experience

Picture yourself embedded in a setting, observing behavior and interactions firsthand. Participant observation involves the researcher becoming part of the environment they are studying. This strategy provides:

- **Rich Contextual Data:** You observe behaviors in their natural setting, gaining insights into the social dynamics and cultural norms that influence them.
- **Understanding Social Interactions:** By observing how people interact, you can uncover unspoken rules, communication patterns, and group dynamics.

Example: You're researching the workplace culture of a start-up company. You might spend a week observing team meetings, interactions in the common area, and participation in social events. This allows you to observe how communication flows, how decisions are made, and the overall atmosphere of the work environment.

4. Focus Group Discussions: A Symphony of Perspectives

Imagine a moderated group discussion where a diverse range of voices come together. Focus groups involve 6-10 participants recruited based on specific criteria. A moderator guides the conversation around a central topic, encouraging everyone to share their perspectives. This approach is ideal for:

- **Generating a Range of Perspectives:** Group discussions can reveal shared ideas, differing viewpoints, and interesting clashes of opinions.
- **Exploring Existing Attitudes:** By facilitating a conversation, you can uncover commonly held beliefs and attitudes about a specific topic.
- **Understanding Group Dynamics:** Focus groups allow you to observe how participants build on each other's ideas, challenge each other's perspectives, and reach a collective understanding.

Example: You're researching the perception of electric vehicles among potential buyers. Your focus group might consist of individuals with varying interests in sustainability, technology, and car ownership. The discussion could reveal concerns about charging infrastructure, range anxiety, or the environmental benefits of electric vehicles.

Choosing the Right Tool:

The most effective data collection strategy depends on your research question and the type of data you need. Consider these factors when making your choice:

- **In-depth exploration of individual experiences:** Semi-structured interviews.
- **Uncovering unforeseen topics and rich narratives:** Unstructured interviews.
- **Understanding social interactions and cultural norms:** Participant observation.
- **Generating a range of perspectives and exploring existing attitudes:** Focus group discussions.

3.4 Coding and thematic analysis

What is coding in qualitative research?

Coding is the process of labeling and organizing your qualitative data to identify different themes and the relationships between them.

When coding customer feedback, you assign labels to words or phrases that represent important (and recurring) themes in each response. These labels can be words, phrases, or numbers; we recommend using words or short phrases, since they're easier to remember, skim, and organize.

Coding qualitative research to find common themes and concepts is part of thematic analysis.

Thematic analysis extracts themes from text by analyzing the word and sentence structure. Within the context of customer feedback, it's important to understand the many different types of qualitative feedback a business can collect, such as open-ended surveys, social media comments, reviews & more

What is qualitative data analysis?

Qualitative data analysis is the process of examining and interpreting qualitative data to understand what it represents.

Qualitative data is defined as any non-numerical and unstructured data; when looking at customer feedback, qualitative data usually refers to any verbatim or text-based feedback such as reviews, open-ended responses in surveys, complaints, chat messages, customer interviews, case notes or social media posts

For example, NPS metric can be strictly quantitative, but when you ask customers why they gave you a rating a score, you will need qualitative data analysis methods in place to understand the comments that customers leave alongside numerical responses.

Methods of qualitative data analysis

1. Content analysis: This refers to the categorization, tagging and thematic analysis of qualitative data. This can include combining the results of the analysis with behavioural data for deeper insights.
2. Narrative analysis: Some qualitative data, such as interviews or field notes may contain a story. For example, the process of choosing a product, using it, evaluating its quality and decision to buy or not buy this product next time. Narrative analysis helps understand the underlying events and their effect on the overall outcome.
3. Discourse analysis: This refers to analysis of what people say in social and cultural context. It's particularly useful when your focus is on building or strengthening a brand.
4. Framework analysis: When performing qualitative data analysis, it is useful to have a framework. A code frame (a hierarchical set of themes used in coding qualitative data) is an example of such framework.
5. Grounded theory: This method of analysis starts by formulating a theory around a single data case. Therefore, the theory is "grounded' in actual data. Then additional cases can be examined to see if they are relevant and can add to the original theory.

Automatic coding software

Advances in natural language processing & machine learning have made it possible to automate the analysis of qualitative data, in particular content and framework analysis

While manual human analysis is still popular due to its perceived high accuracy, automating the analysis is quickly becoming the preferred choice. Unlike manual analysis, which is prone to bias and doesn't scale to the amount of qualitative data that is generated today, automating analysis is not only more consistent and therefore can be more accurate, but can also save a ton of time, and therefore money.

3.5 Types of qualitative research

3.5.1 Ethnography

3.5.2 Narrative study

1. **Descriptive Methodology**:
    - Description: Descriptive qualitative research aims to systematically describe a phenomenon, situation, or problem.
    - Descriptive qualitative research aims to systematically describe a phenomenon, situation, or problem. It focuses on understanding the "who, what, and where" of events or experiences and provides insights into poorly understood phenomena. It often utilizes in-depth interviews for data collection and emphasizes rich descriptive summaries of the characteristics of the phenomenon.
    - Example: A study by Cao et al. (2022) explored the state of education regarding end-of-life care from the perspectives of undergraduate nurses. The study provided a rich descriptive summary of the characteristics of the undergraduate curriculum related to end-of-life care and the cultural attitudes toward disease and death that impede nurses' learning and knowledge translation.
2. **Phenomenology**:
    - Description: Phenomenology seeks to understand the essence of a particular phenomenon through a detailed exploration of individual experiences.
    - It focuses on personal experiences such as emotions, perceptions, and awareness. Phenomenological research involves in-depth conversations with participants and employs data analysis techniques to identify the essential structure or meaning of the experience being studied.
    - Example: Liao et al. (2021) conducted a study exploring medical learners' experiences through narrative medicine and the meanings they ascribe to narrative-based learning. The study identified themes such as feeling hesitation, seeking guidance, and experiencing transformation, providing

insights into the essence of medical learners' experiences with narrative-based learning.

3. **Narrative Inquiry**:
    - Description:
    - Narrative inquiry explores how individuals make meaning of their lives and the world around them through studying their stories and experiences. It focuses on giving voice to marginalized populations and understanding the relationship between personal experience and the wider social and cultural context. Narrative inquiry involves collecting and analyzing stories or narratives to identify themes, patterns, and meanings.
    - Example: Gordon et al. (2015) conducted a narrative inquiry exploring medical trainees' experiences of leadership and followership in the interprofessional healthcare workplace. The study revealed themes related to the factors facilitating or inhibiting the development of leadership identities among medical trainees, providing valuable insights into their lived experiences.
4. **Case Study**:
    - Description:
    - Case study methodology provides a holistic exploration of a phenomenon within its real-life context. It offers detailed accounts of individual cases to understand complex phenomena and explore new research questions. Case studies can be intrinsic, instrumental, or collective and involve data collection from various sources such as interviews, observations, and documents.
    - Example: Lemmen et al. (2021) conducted a case study to provide insight into how adopting positive health (PH) in a general practice affects primary care professionals' (PCP) job satisfaction. The study identified themes related to PCPs' adoption of PH, changes in organizational structures, and increased job satisfaction, offering a detailed account of the phenomenon within its real-life context.
5. **Ethnography**:
    - Description: Ethnography involves the study of culture and everyday life within natural settings. It focuses on understanding the lived culture of a group of people and typically involves direct observation of participants over time. Ethnographic research aims to capture the richness and complexity of social phenomena within their cultural context.
    - Example: Hinder and Greenhalgh (2012) conducted an ethnographic study to produce a richer understanding of how people live with diabetes and why self-management is challenging for some. The study revealed insights into the socio-cultural and environmental factors influencing individuals' self-management practices and highlighted the complexities of living with diabetes within the context of everyday life.
6. **Action Research**:
    - Description: Action research is a cyclical process of planning, action, observation, and reflection aimed at improving practice or addressing a problem. It involves

collaborative inquiry between researchers and participants to understand and improve complex social systems. Action research often employs qualitative methods such as interviews and observation to generate new knowledge and understanding.
- Example: Doherty and O'Brien (2021) conducted action research to explore midwives' understandings of burnout in the context of contemporary maternity care in Ireland. The study involved cycles of action and reflection, leading to insights into midwives' views on burnout and collaborative efforts to develop and maintain change within the healthcare system.

7. **Grounded Theory**:
    - Description: Grounded theory is a qualitative research approach that emphasizes developing theories based on evidence collected from participants. It involves inductive data analysis to identify patterns and relationships in the data and build hypotheses and insights based on the findings. Grounded theory is particularly valuable when little is known about a phenomenon and aims to develop explanatory theories grounded in the data.
    - Example: Malau-Aduli et al. (2020) conducted a grounded theory study to identify factors influencing International Medical Graduates' (IMG) decision to remain working in regional, rural, and remote (RRR) areas. The study developed a theory grounded in data, explaining how IMGs prioritize and evaluate factors influencing their decision to work in RRR areas.

Each of these qualitative methodologies offers unique strengths and approaches to